



وزارة الاتصالات
وتكنولوجيا المعلومات



Excel 1



Excel



What is excel?

- Excel is a spreadsheet program from Microsoft and a component of its Office product group for business applications. Microsoft Excel enables users to format, organize and calculate data in a spreadsheet.



Who use excel?

- collection and verification of business data
- business analysis
- data entry and storage
- data analysis
- performance reporting
- strategic analysis
- accounting and budgeting
- administrative and managerial management
- account management
- office administration



Let's start with the excel overview



Excel

Home

New

Open

Account

Feedback

Options

Good morning

New

Blank workbook

Monthly company budget

Make your first PivotTable

Sales invoice tracker

Welcome to Excel

Get started with Formulas

Gantt project planner

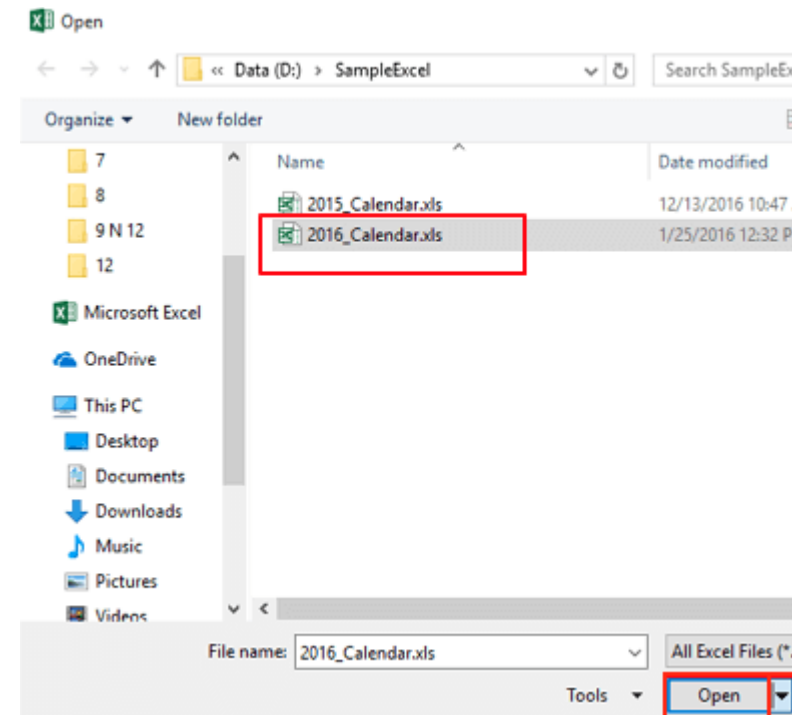
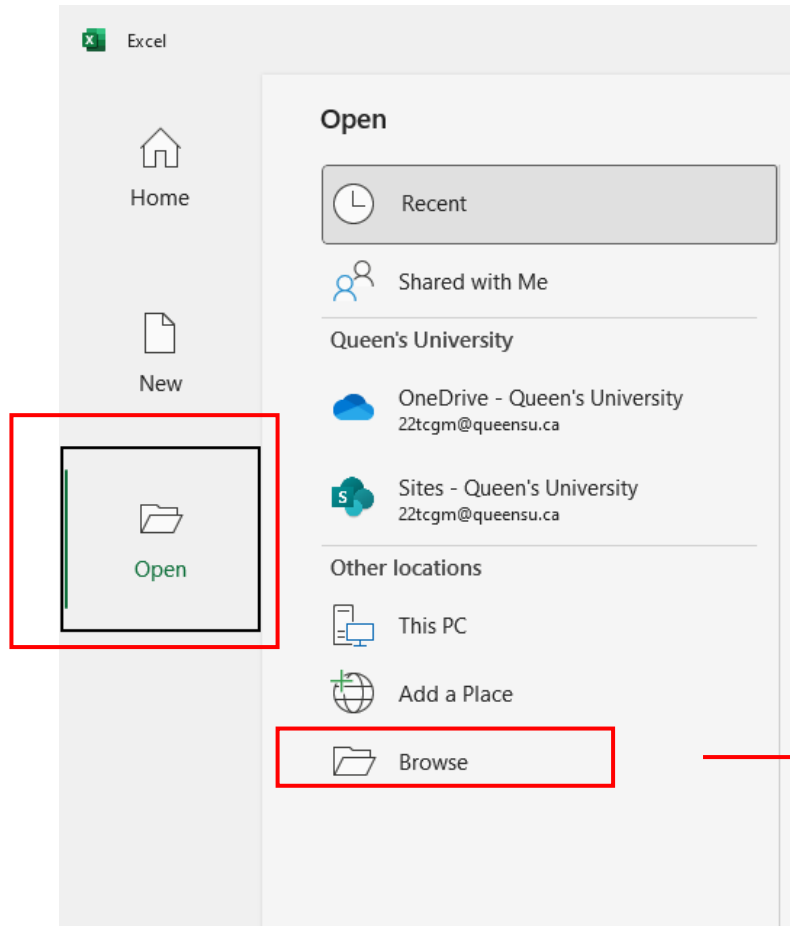
More templates →

Search

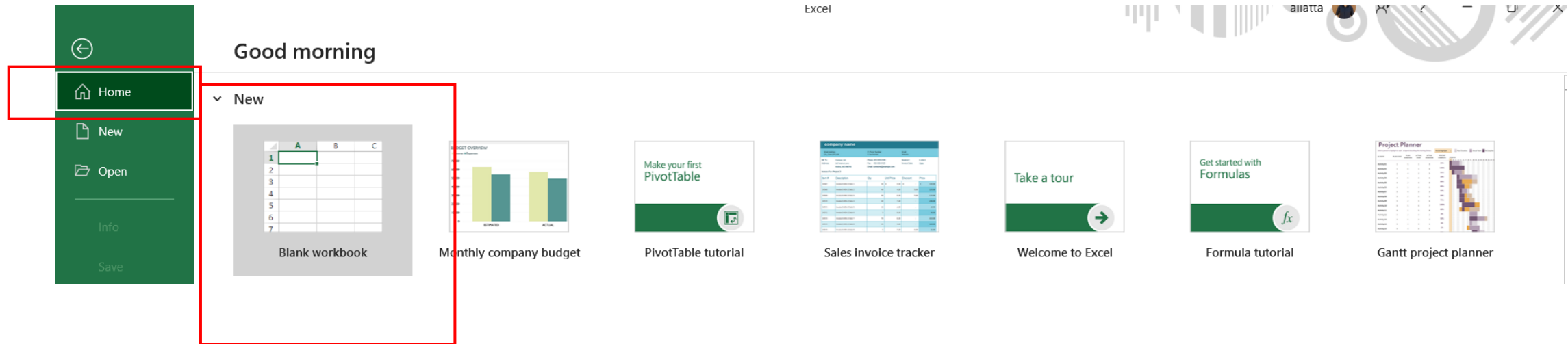
Recent Pinned Shared with Me

	Name	Date modified
	Book1 OneDrive - Beni-suef University faculty of fcis	Thu at 1:11 AM
	pandas32 Desktop » New folder	Tue at 12:30 AM
	data Desktop	Tue at 12:09 AM
	SampleSuperstore D: » abdel	Mon at 9:49 PM
	rabee Desktop » New folder	Mon at 9:45 PM
	ali Desktop » ali	Sun at 3:52 AM

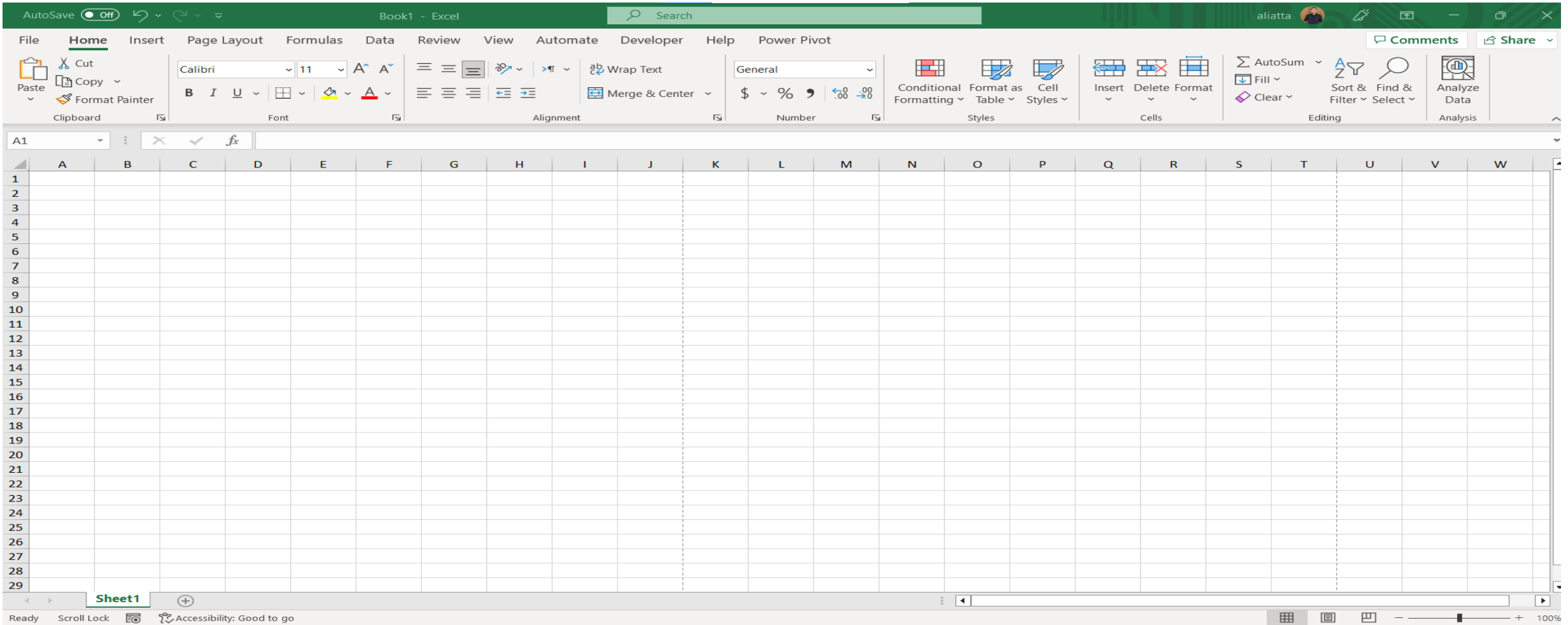
How to open existing file



How to open a new file?



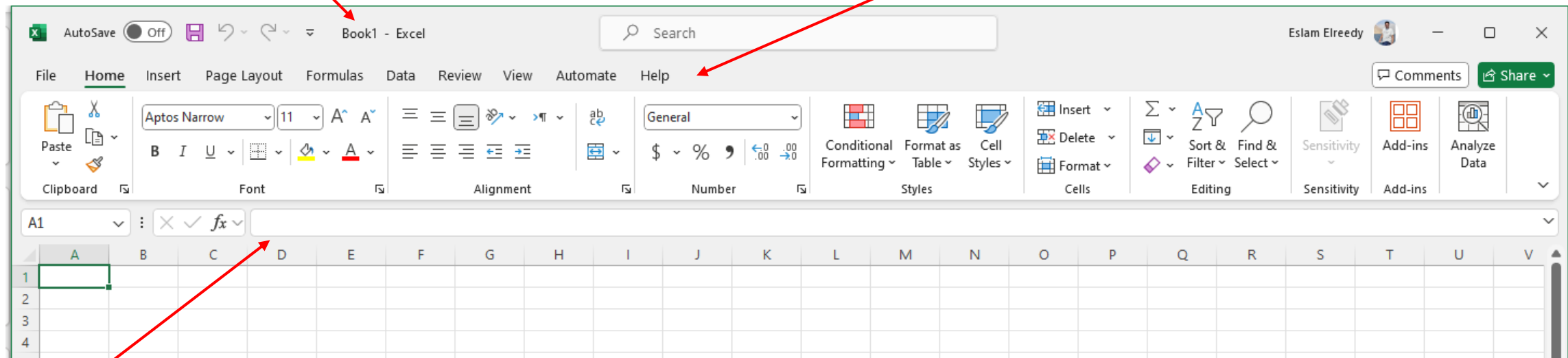
Look what will happen



Overview with the interface

Title Bar: Displays the name of the current workbook .

Ribbon: Contains tabs (Home, Insert, Page Layout, etc.) with groups of related commands (Font, Alignment, Number, etc.).

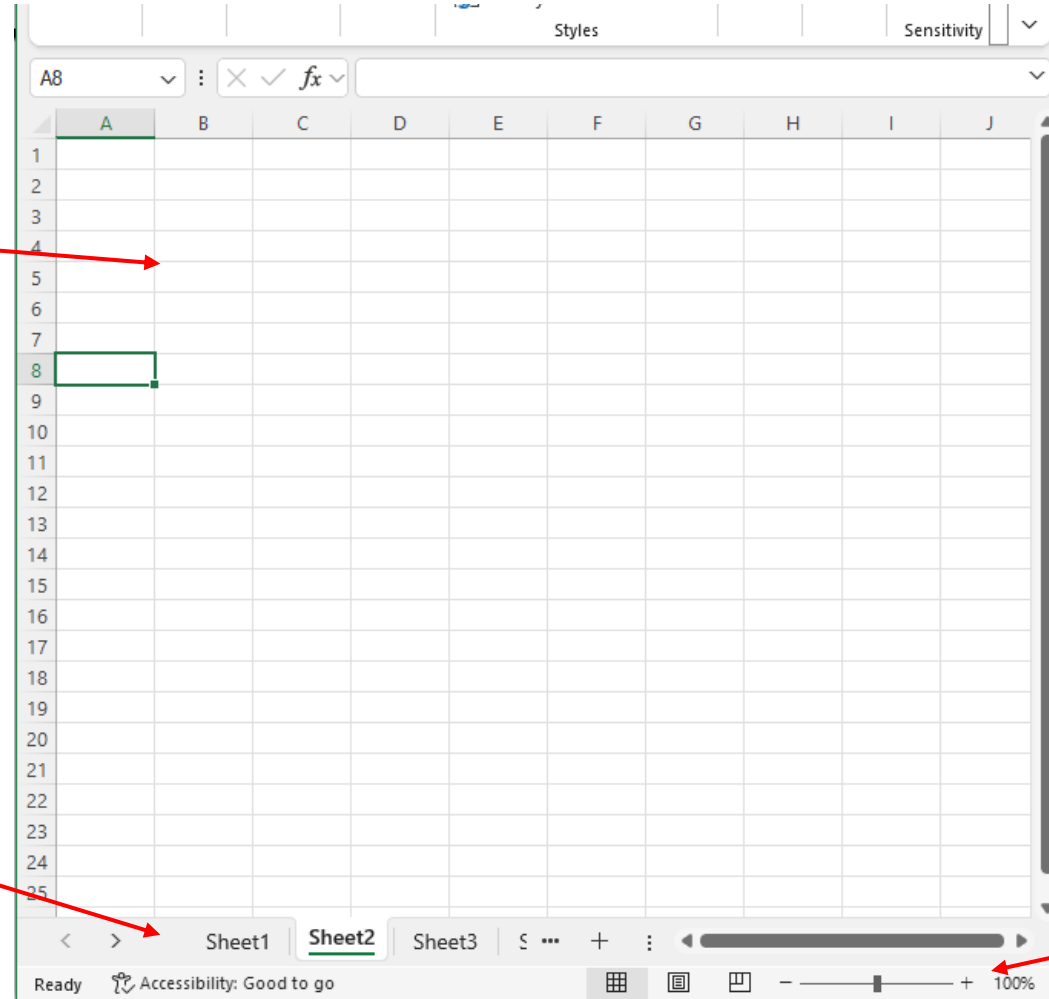


Formula Bar: Shows the content of the currently selected cell and allows for formula entry and editing.

Overview with the interface

Worksheet Area: The grid where data is entered. Each cell is identified by a column letter and row number.

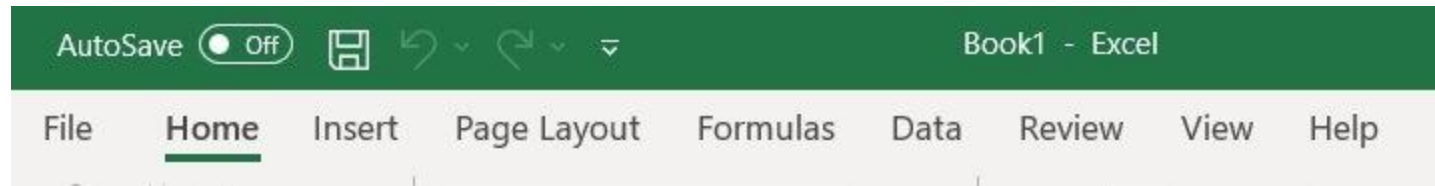
Sheet Tabs: Located at the bottom of the window, allowing navigation between different worksheets within a workbook.



Status Bar: Displays information about the current status of the workbook and includes shortcuts for viewing mode and zoom level.

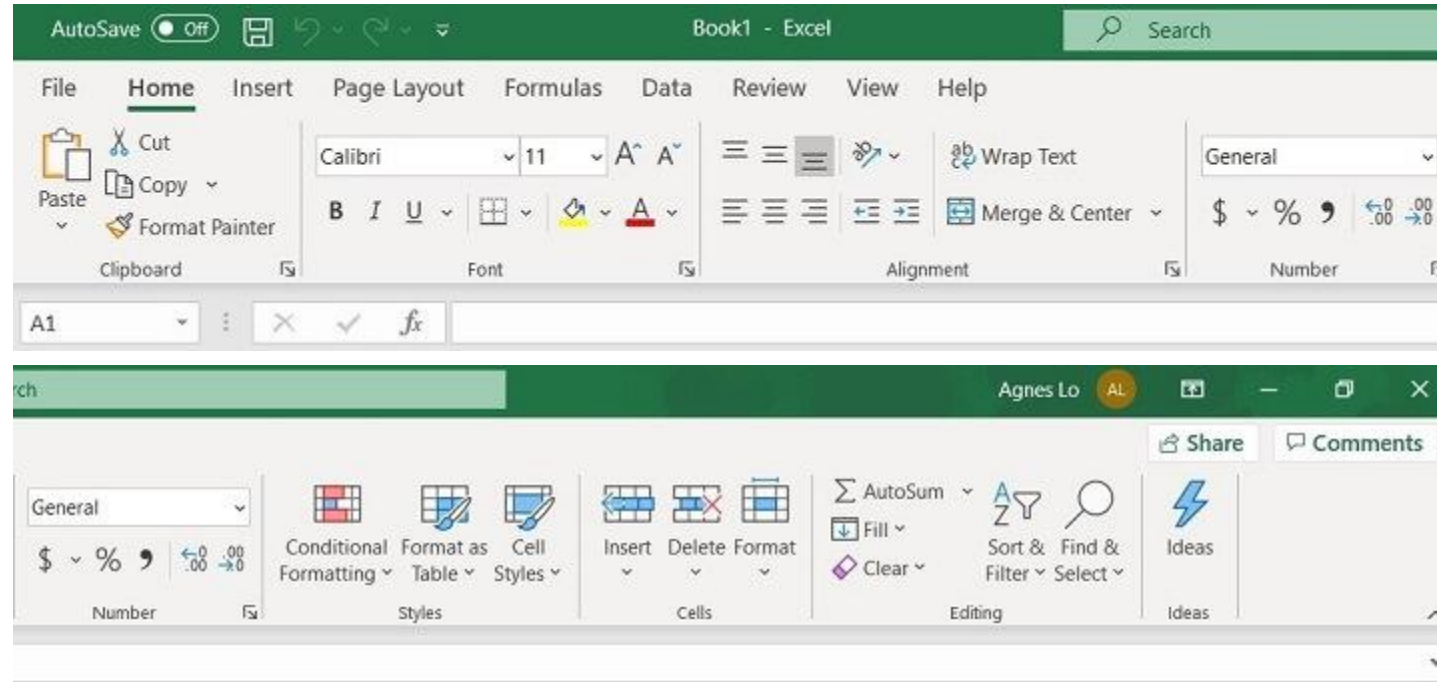
The Excel ribbon tabs

- There are nine tabs on the Excel Ribbon: **File, Home, Insert, Page Layout, Formulas, Data, Review, View, and Help**. The Home tab is the default tab when Excel is opened.



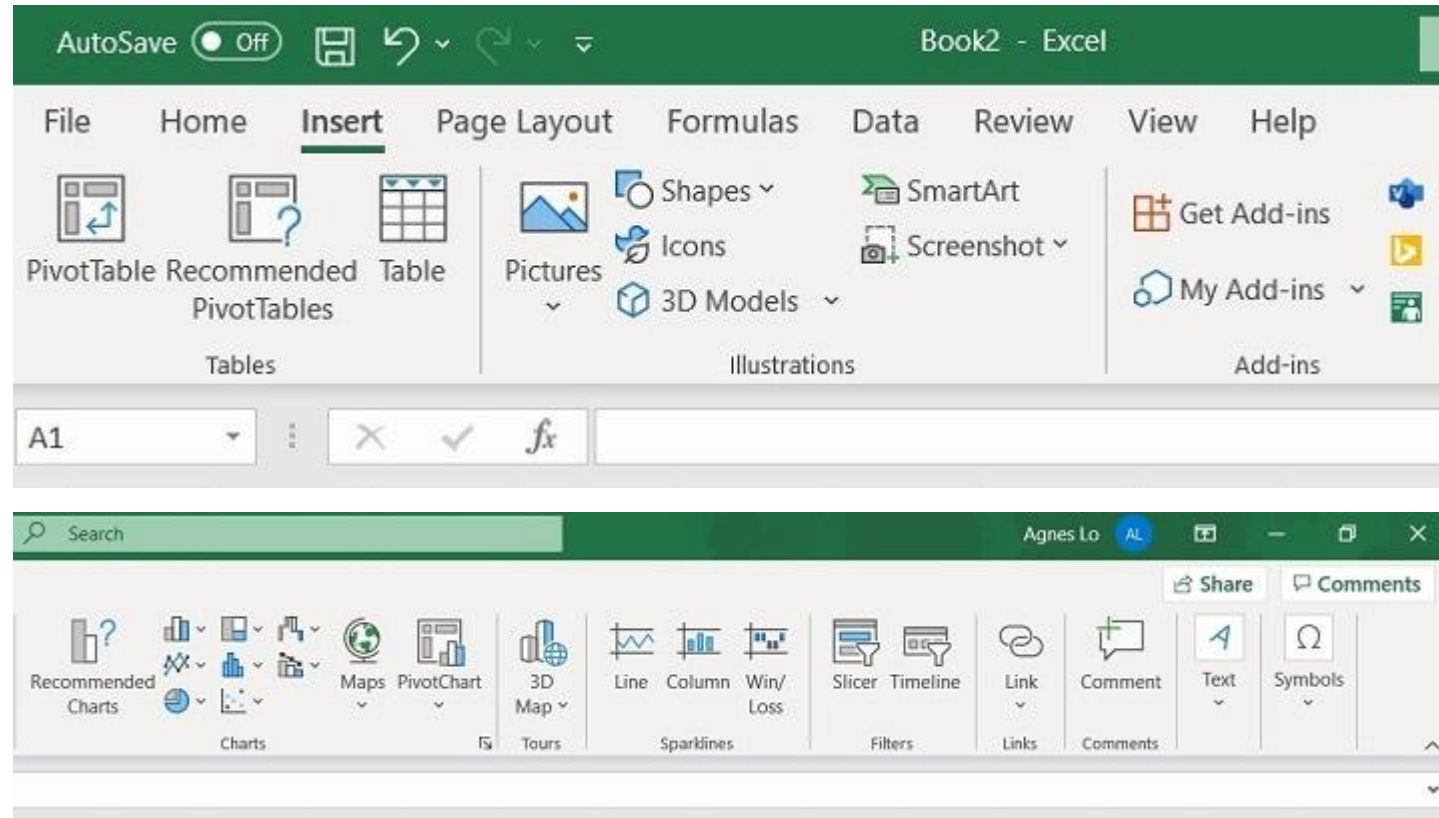
Home

- Contains common formatting tools, such as font style, alignment, and number formatting, along with basic functions like copy, paste, and sorting



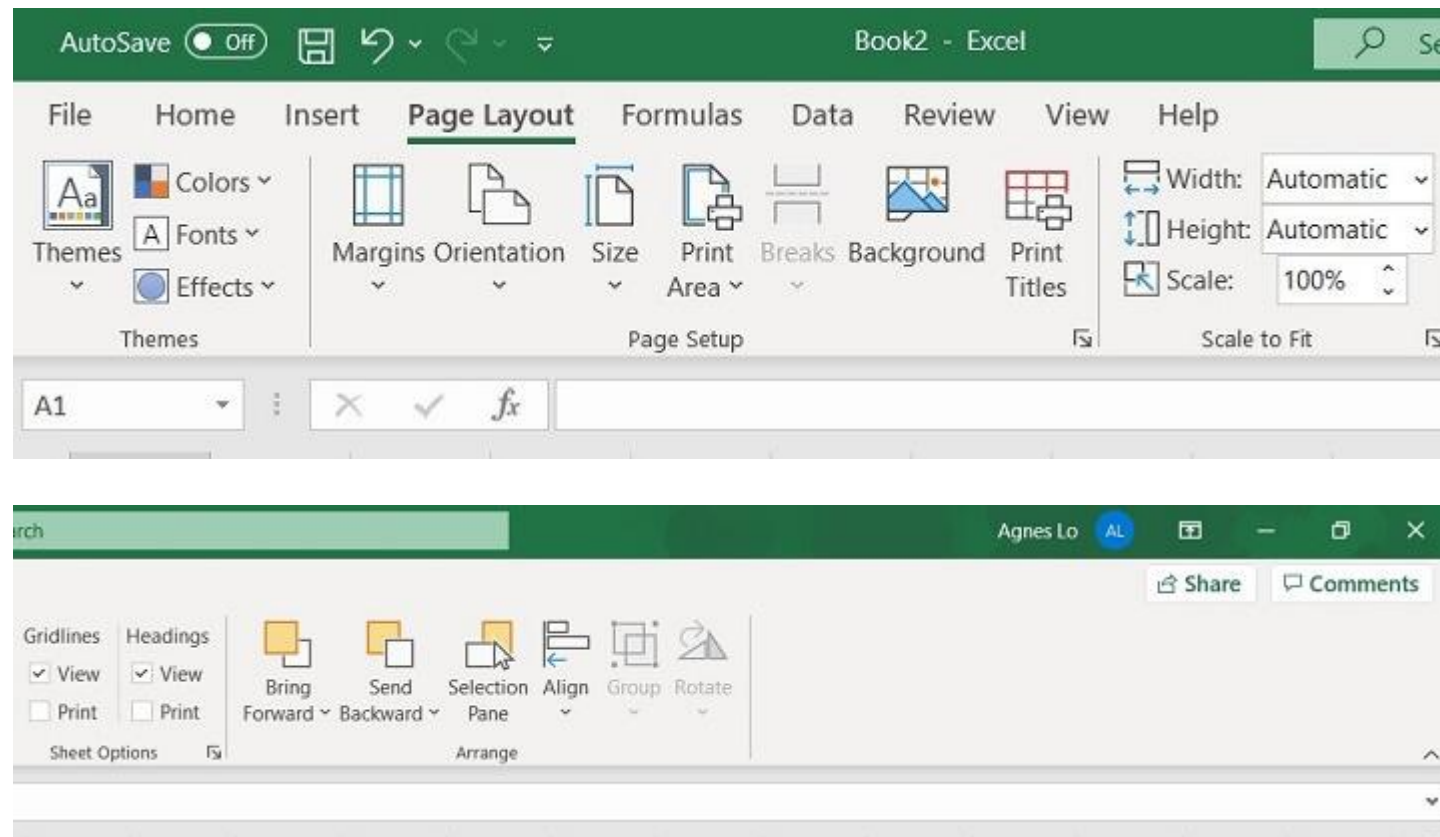
Insert

This is where users can add various items to a spreadsheet, including Pivot Tables, pictures, shapes, charts, graphs, and symbols.



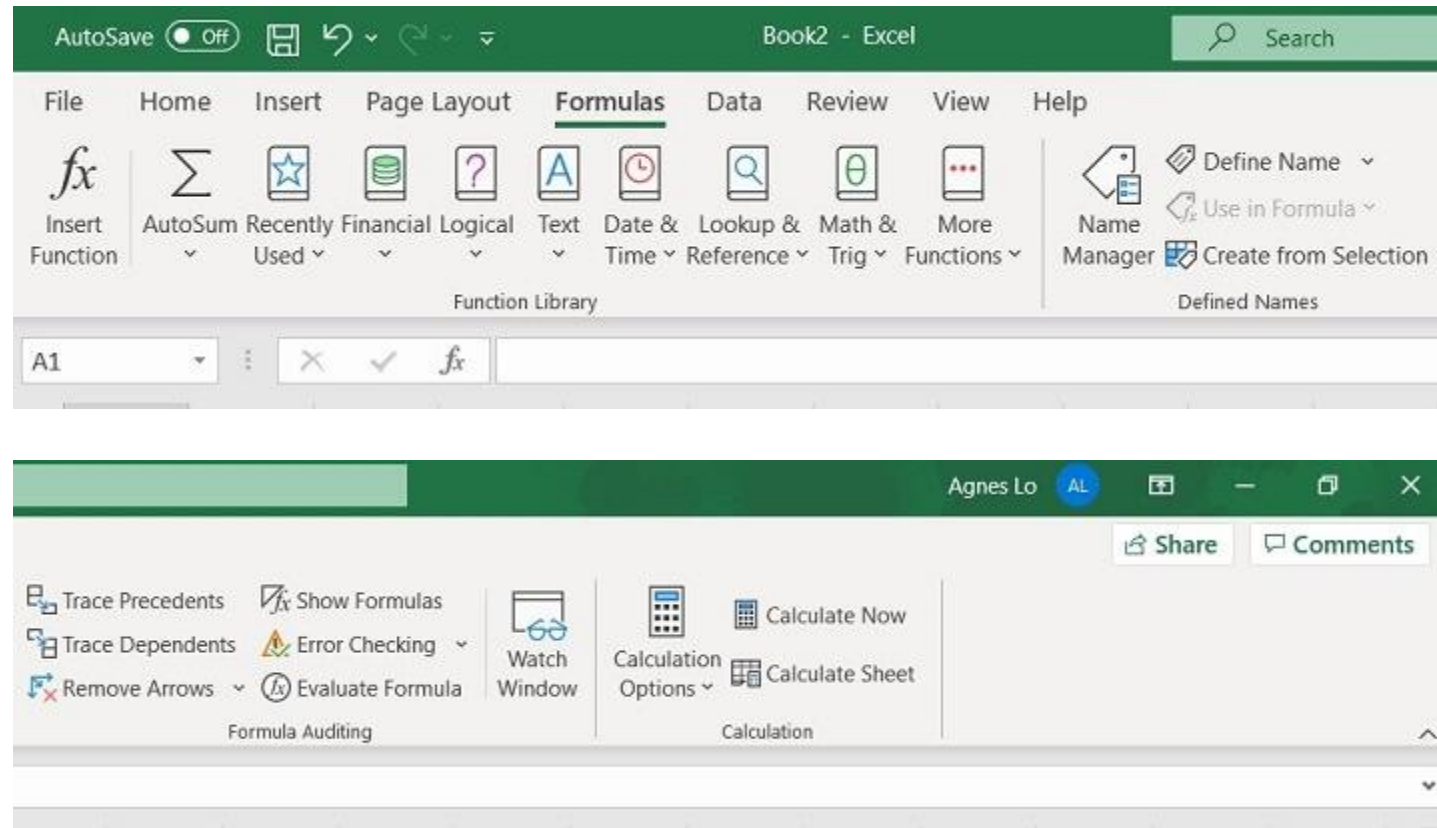
Page Layout

This allows users to customize the layout of the spreadsheet by adjusting the margins, color themes, gridlines, and print area. The changes are applicable when being printed as well.



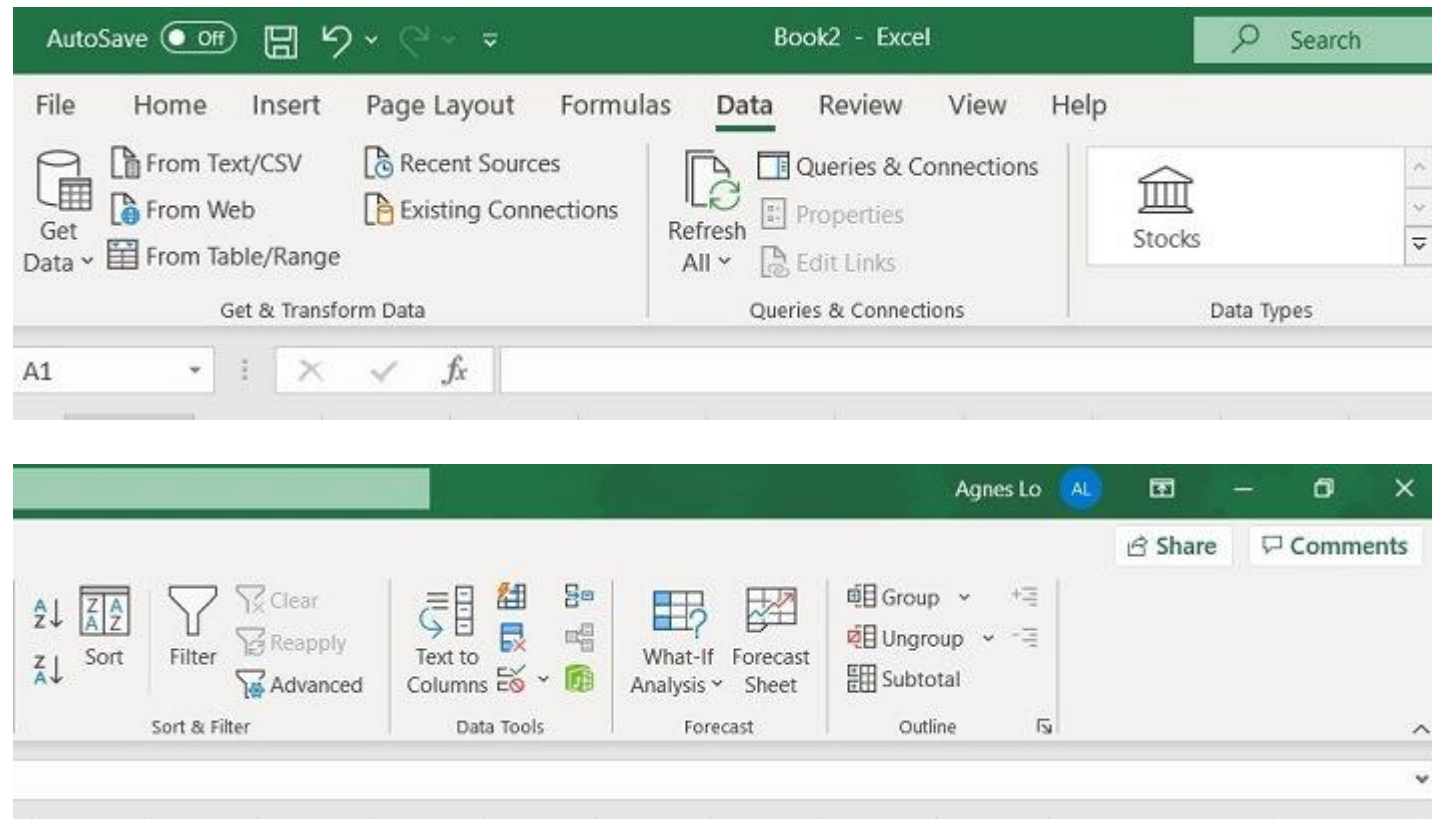
Formulas

This is where all essential formulas are categorized under the function library, and it provides various controlling options.



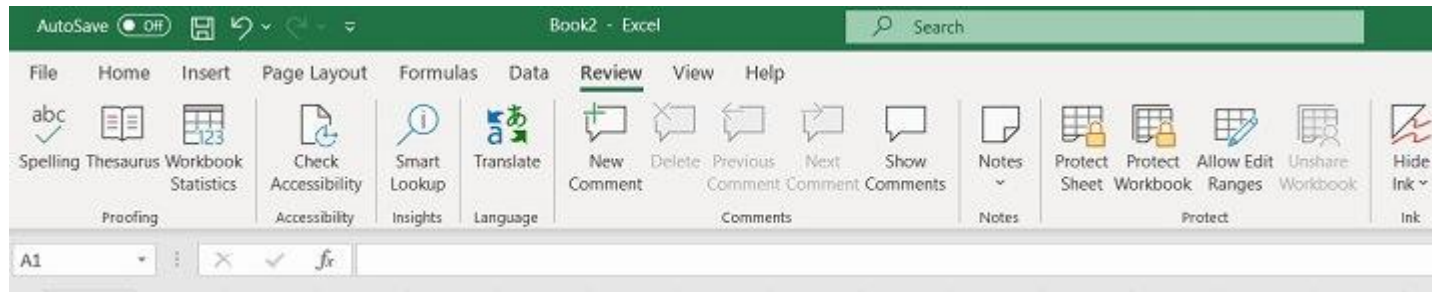
Data

This allows users to manage data in the current spreadsheet, within the file, and also to import external data from other sources.



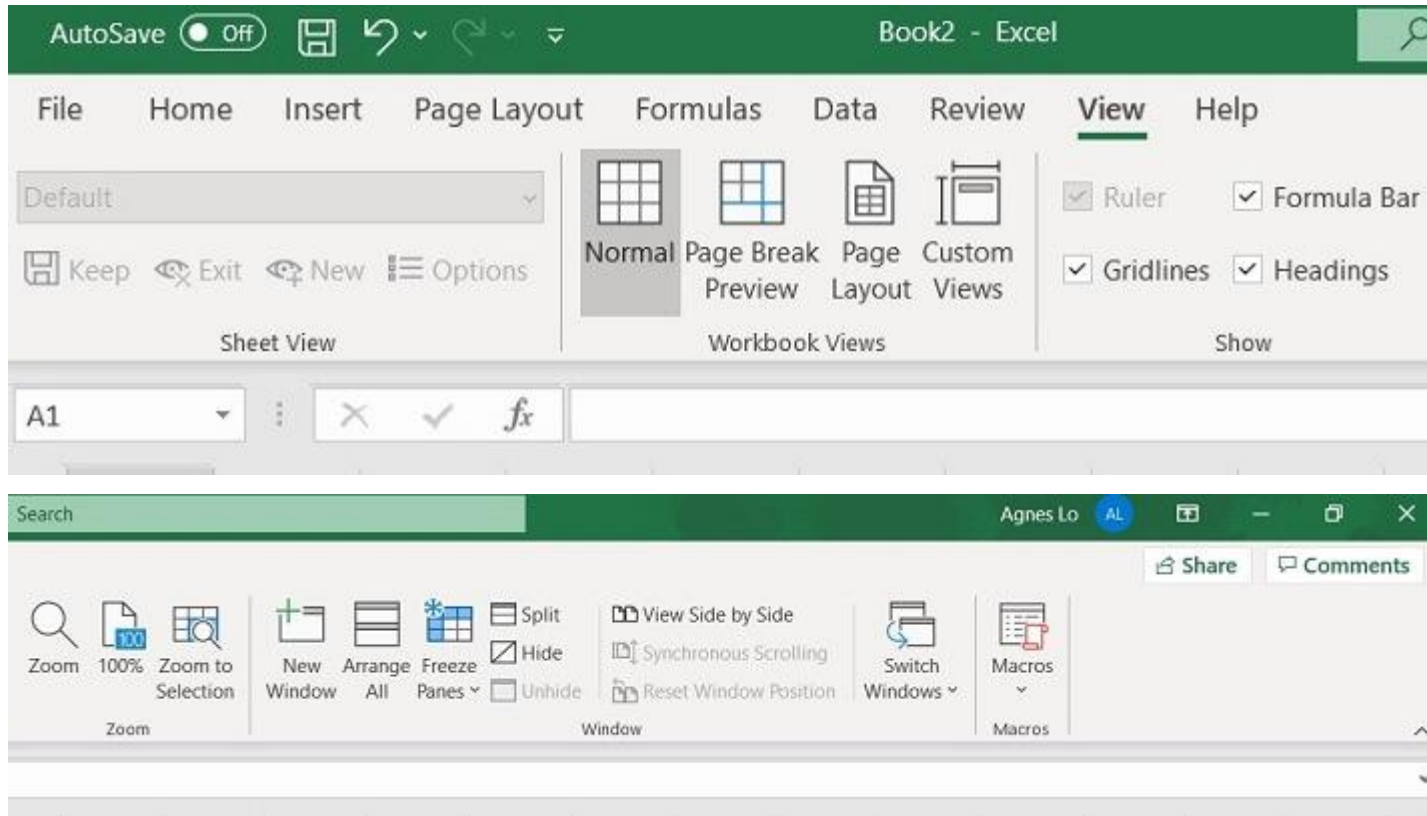
Review

This is for users to perform various controlling functions, including spell check, translate, adding comments and notes, track changes, and to enable worksheet protection.



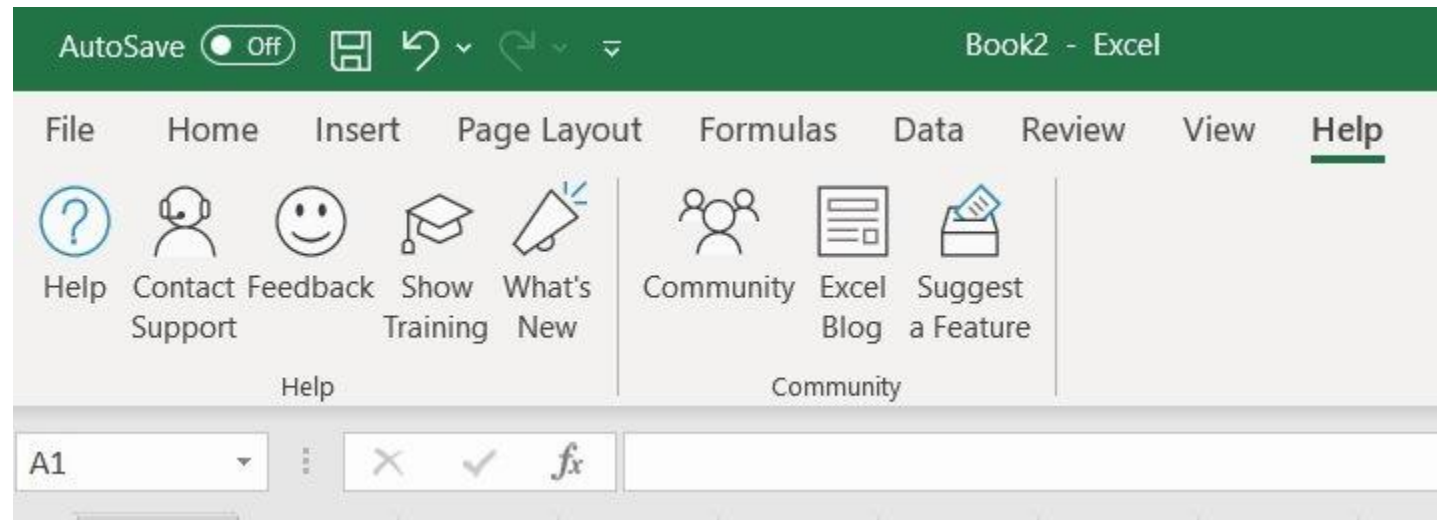
View

This provides options to alter the views of worksheets — gridlines, zoom, freeze panes, and to switch windows.



Help

This gives access to [Microsoft support](#). It allows you to provide feedback and to suggest a feature to the community.

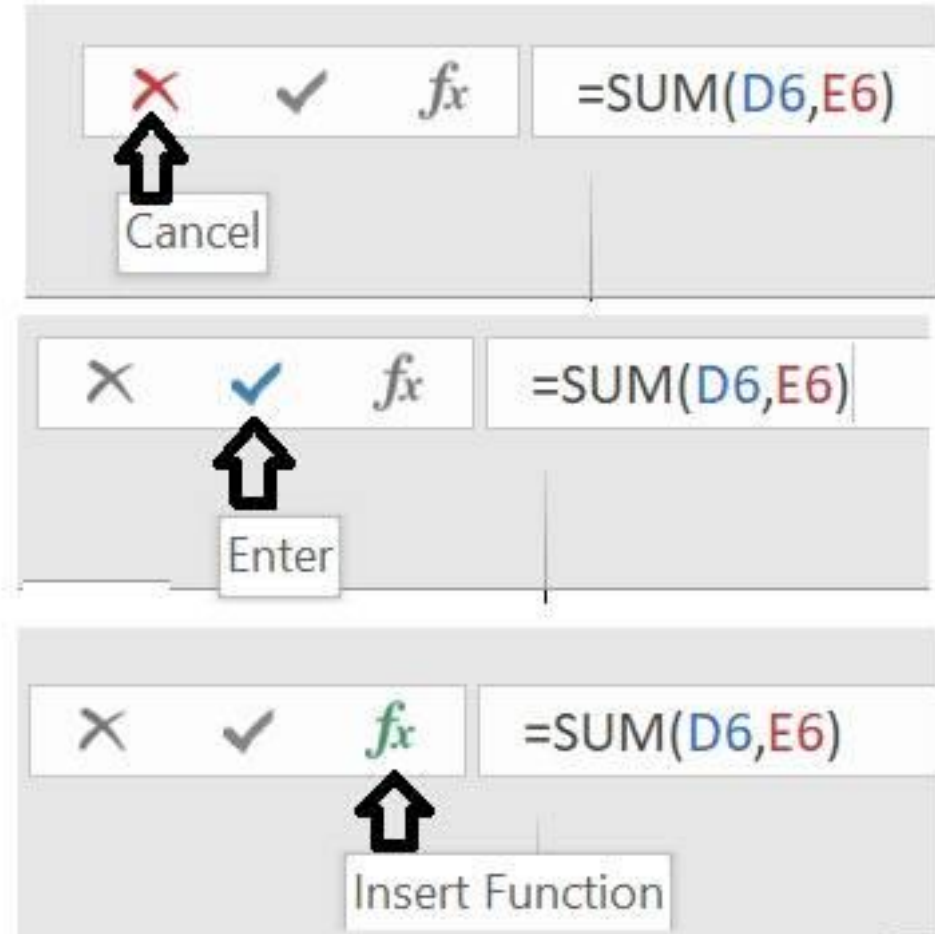


What is the Formula Bar in Excel?

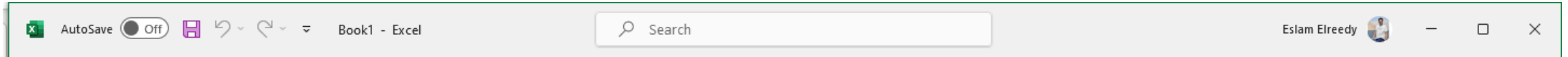
- The Excel formula bar is the wide rectangular field between the ribbon and the spreadsheet cell grid and to the right of the name box. It is the first element we use to define the mathematical calculation. The formula bar in Excel is the space where the values or expressions that will perform the mathematical operations appear.
- The formula bar makes it easy to create and modify formulas and provides information about functions and their arguments.

Parts of the Excel Formula Bar

- The Excel formula bar consists of three main parts:
- **Name Box:**
 - **Location:** Left side of the formula bar.
 - **Function:** Displays the name of the selected cell or range. It can also show the name of a function being used.
- **Function Wizard:**
 - **Function:** Allows you to use built-in functions to operate on spreadsheet data.
 - **Components:**
 - **Cancel (X):** Cancels an ongoing operation.
 - **Enter (Check Mark):** Confirms the implementation of a function.
 - **Insert Function (fx):** Opens a dialogue to select a function and enter its arguments more easily.



Components of the Excel Title Bar

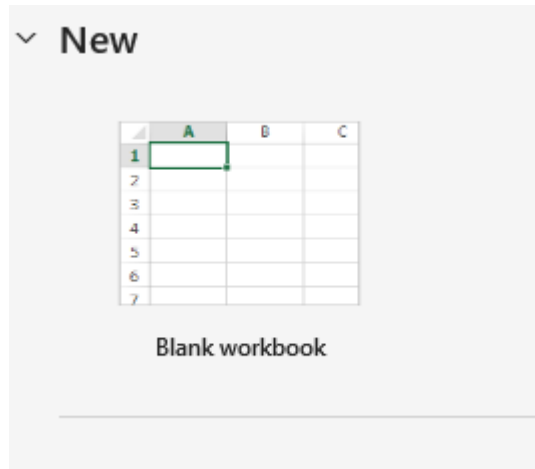


- **1. Workbook name:** The title bar displays the name of the active excel workbook in front and centre. This helps users in instant identification, though they are always aware of the file working on, even amongst a multitude of open spreadsheets.
- **Window controls:** The window controls include the familiar trio like:
 - Minimise
 - Maximise/ restore
 - Close buttons
- **Quick access toolbar:** This customizable toolbar contains frequently used commands for swift access and is nested at the left end. This personalization empowers users to tailor their excel workspace to their needs in order to boost efficiency.

Workbook and Worksheet Concepts

Workbook:

- A workbook is a file that contains one or more worksheets.
- Each workbook has a .xlsx extension (or .xls for older versions).
- Workbooks can be used to organize related data in separate sheets.



Worksheet:

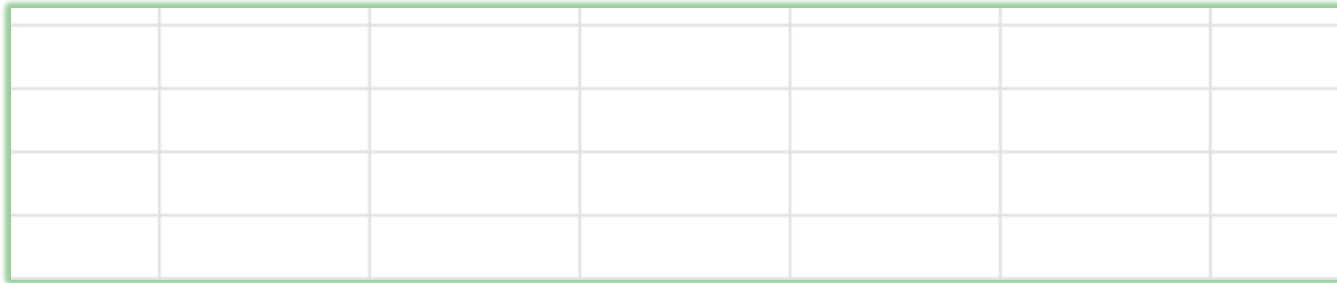
- A worksheet is a single page within a workbook.
- Each worksheet consists of cells organized in rows and columns.
- You can add, rename, and delete worksheets as needed.



THE CELL IN EXCEL



- Cell the intersection of row and column in work sheet



Basic Navigation and Shortcuts

Navigating Cells:

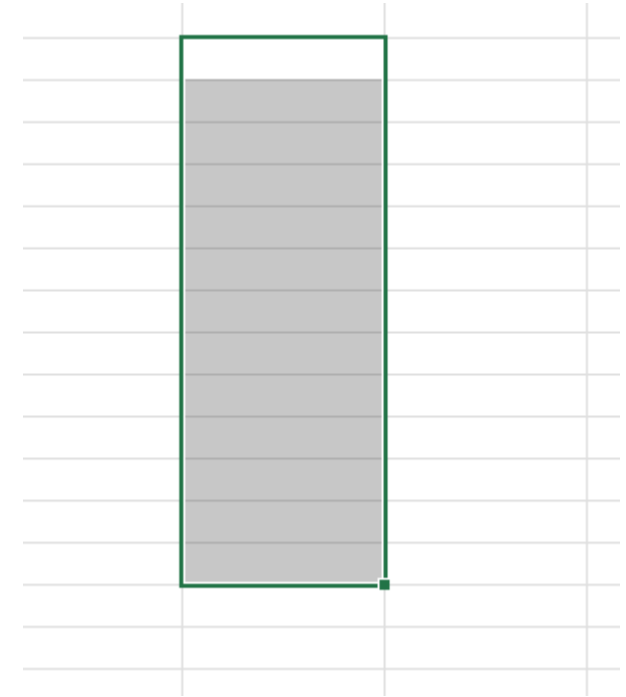
- **Arrow Keys:** Move one cell up, down, left, or right.
- **Enter:** Move one cell down.
- **Tab:** Move one cell right.
- **Home:** Move to the beginning of the row.
- **Ctrl + Arrow Keys:** Jump to the edge of data regions.

THE CELL IN EXCEL



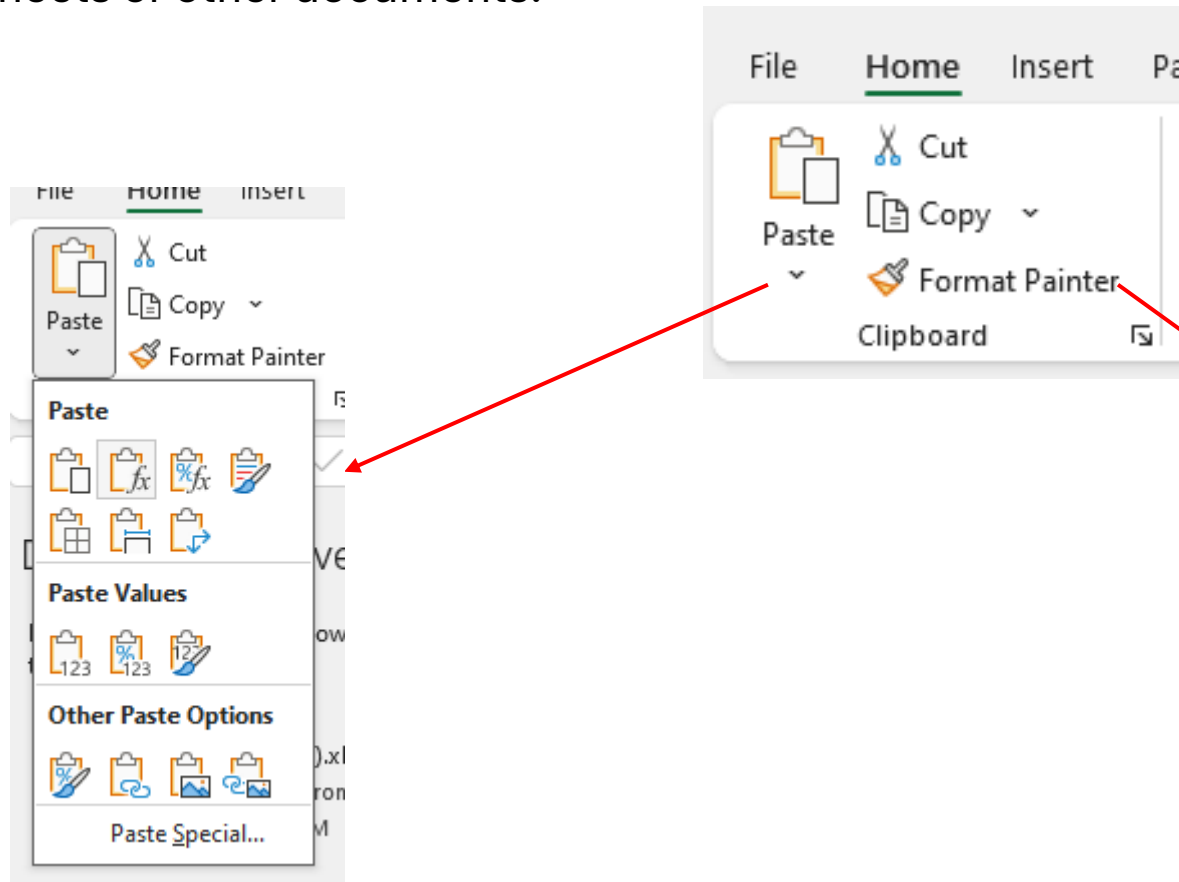
Selecting Cells:

- **Click and Drag:** Select a range of cells by clicking and dragging.
- **Shift + Arrow Keys:** Extend the selection one cell at a time.
- **Ctrl + A:** Select **the entire** worksheet.
- **Ctrl + Shift + Space:** Select **the entire current data region**.



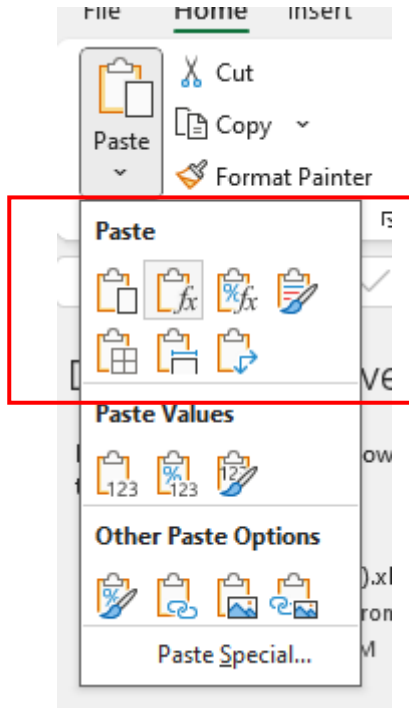
THE CLIPBOARD

allows you to copy and store multiple items (text, graphics, and data) and then paste them into your worksheets or other documents.



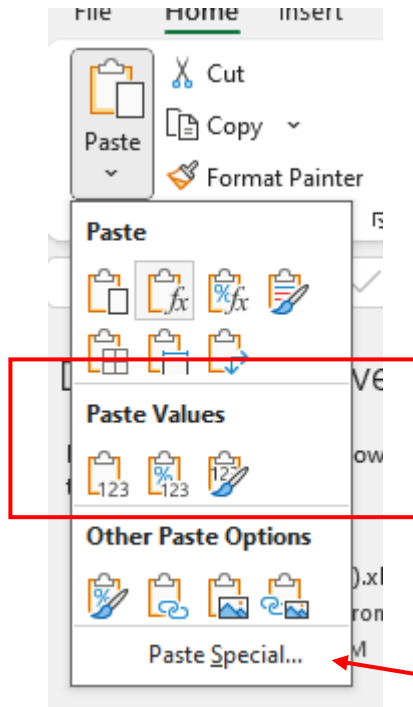
Format Painter allows you to quickly copy formatting from one part of your worksheet to another.

THE CLIPBOARD



- Keep Source Formatting: Retains the formatting of the copied cells.
- Match Destination Formatting: Adopts the formatting of the destination cells.
- Formulas: Pastes only the formulas from the copied cells.
- Formulas and Number Formats: Pastes the formulas and number formatting but not other formatting like font or cell color.
- Keep Source Column Widths: Pastes the data and the column widths from the source.
- No borders: paste without the borders
- Transpose: Changes the orientation of the copied data from rows to columns or vice versa.

THE CLIPBOARD



Values: Pastes only the values, not the formulas.

Values and Number Formats: Pastes values and number formats only.

Values and Source Formatting: Pastes values and retains source formatting.

Paste Special: Opens the Paste Special dialog box for more advanced options.

FONT

click the dropdown arrow next to the current font style to display a list of available styles.

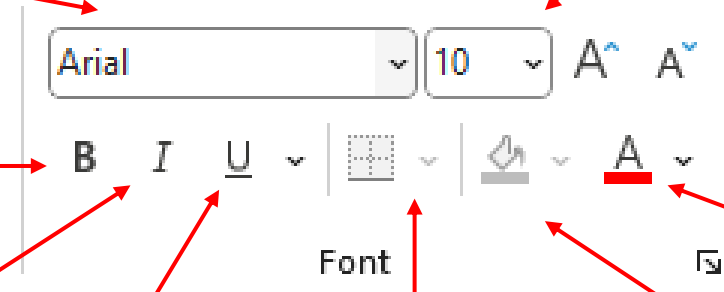
Click the "**B**" button to apply bold.

Click the "**I**" button to apply italic.

Click the "**U**" button to apply underline.

Manage the borders style

click the dropdown arrow next to the current font size to display a list of available sizes.
Select the desired font size from the list or type a custom size directly into the font size box and press Enter.



Color of the text

Color the background of cell to make it solid

Common Shortcuts:



- **Ctrl + C:** Copy selected cells.
- **Ctrl + V:** Paste copied cells.
- **Ctrl + X:** Cut selected cells.
- **Ctrl + Z:** Undo the last action.
- **Ctrl + Y:** Redo the last undone action.

- **Ctrl + F:** Open the Find dialog box.
- **Ctrl + H:** Open the Replace dialog box.
- **F2:** Edit the selected cell's content.
- **Alt + E, S, V:** Paste Special.

A L I G N M E N T

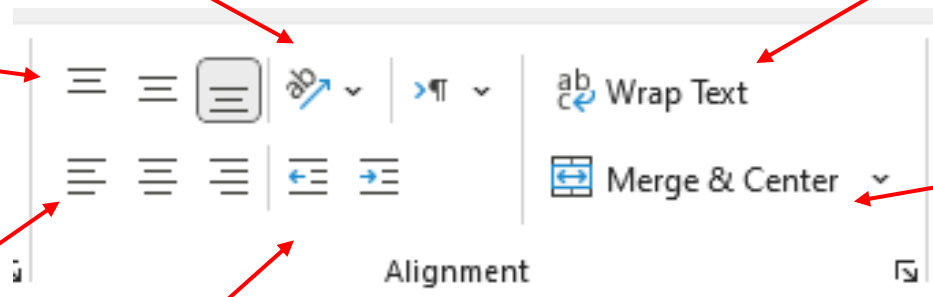


Orientation changes the angle of the text within the cell.

Top Align
Middle Align
Bottom Align

Align Left
Center
Align Right

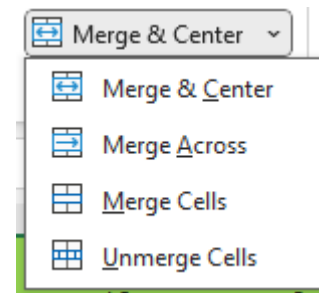
Indent Decrease
Indent Increase



Wrap Text makes all content visible within a cell by displaying it on multiple lines.

Merge & Center combines multiple selected cells into one cell and centers the text.

Merge Across
Merge Cells
Unmerge Cells

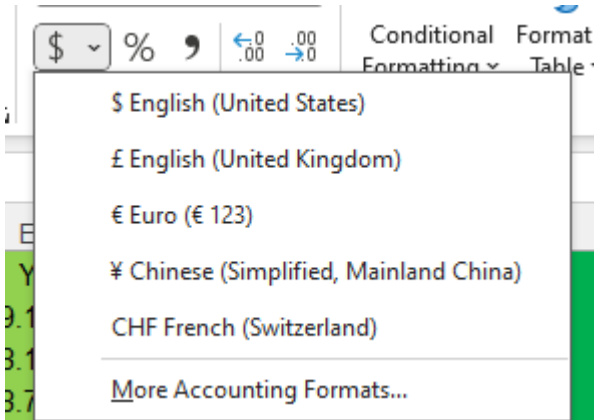


Numbers

It includes the following tools:

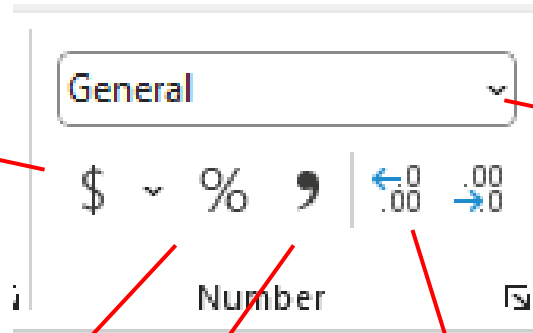


Accounting Number Format



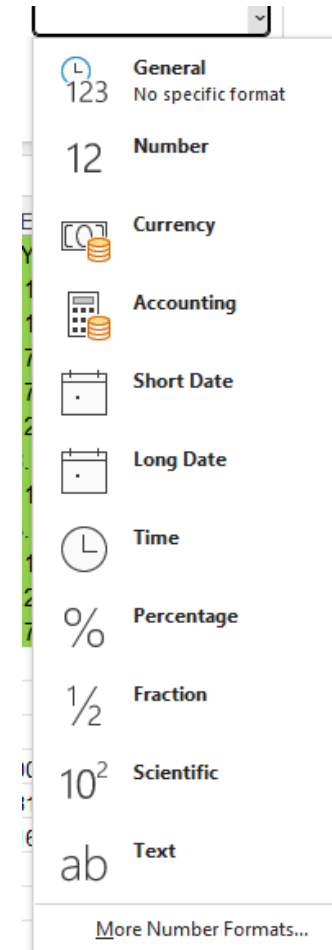
Percent Style

Comma Style: format numbers with thousand separators and two decimal places.

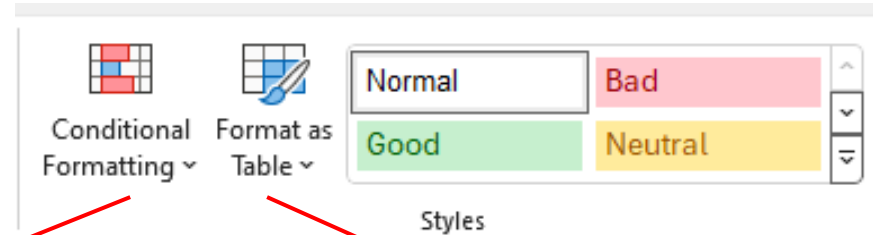


Increase Decimal
Decrease Decimal

Number Format Dropdown



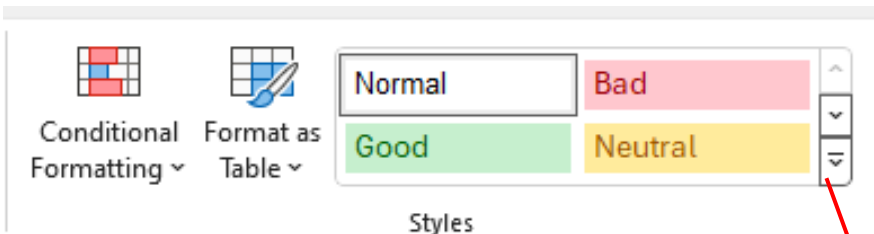
Style



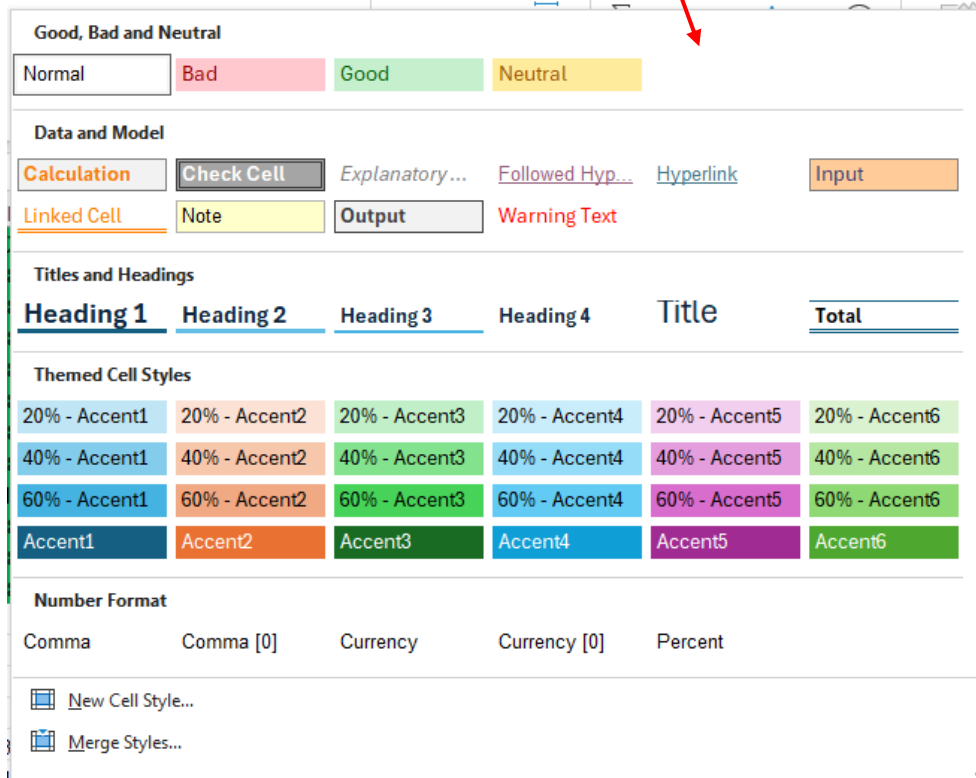
Conditional Formatting changes the appearance of cells based on specific criteria. This is useful for highlighting important information, identifying trends, or making your data easier to interpret.

Format as Table converts a range of cells into a table with a predefined style. This not only changes the appearance but also adds functionality like sorting, filtering, and structured references quickly.

Style



Cell Styles apply predefined formatting options to selected cells, including font, number format, cell borders, and fill color. This is useful for ensuring consistent formatting across your worksheet.



Normal: Default cell style.

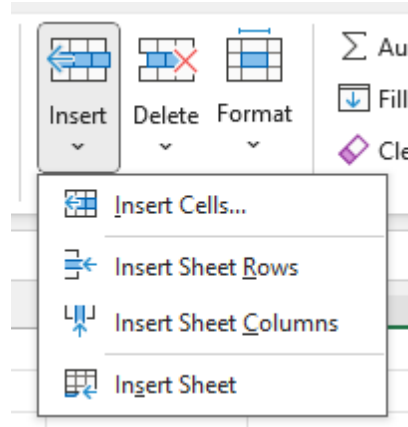
Bad, Good, Neutral: Highlight cells as good, bad, or neutral.

Titles and Headings: Apply styles to titles and headings.

Themed Cell Styles: Apply styles that match the current theme.

Number Format: Apply specific number formats (e.g., Currency, Percent).

cells



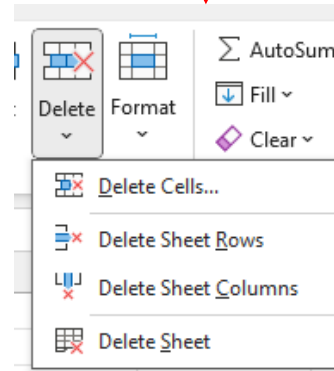
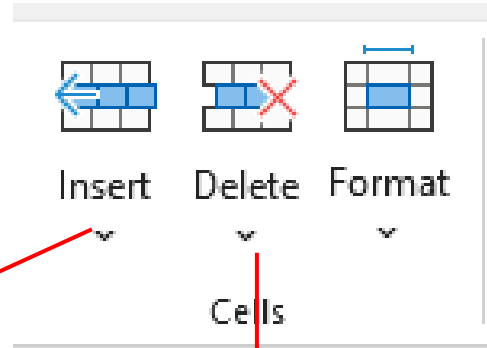
Insert allows you to add new cells, rows, columns, and sheets to your workbook.

Insert Cells: Adds new cells at the selected location, shifting existing cells down or to the right.

Insert Sheet Rows: Adds new rows above the selected row(s).

Insert Sheet Columns: Adds new columns to the left of the selected column(s).

Insert Sheet: Adds a new worksheet to the workbook.



Delete removes cells, rows, columns, or sheets from your workbook.

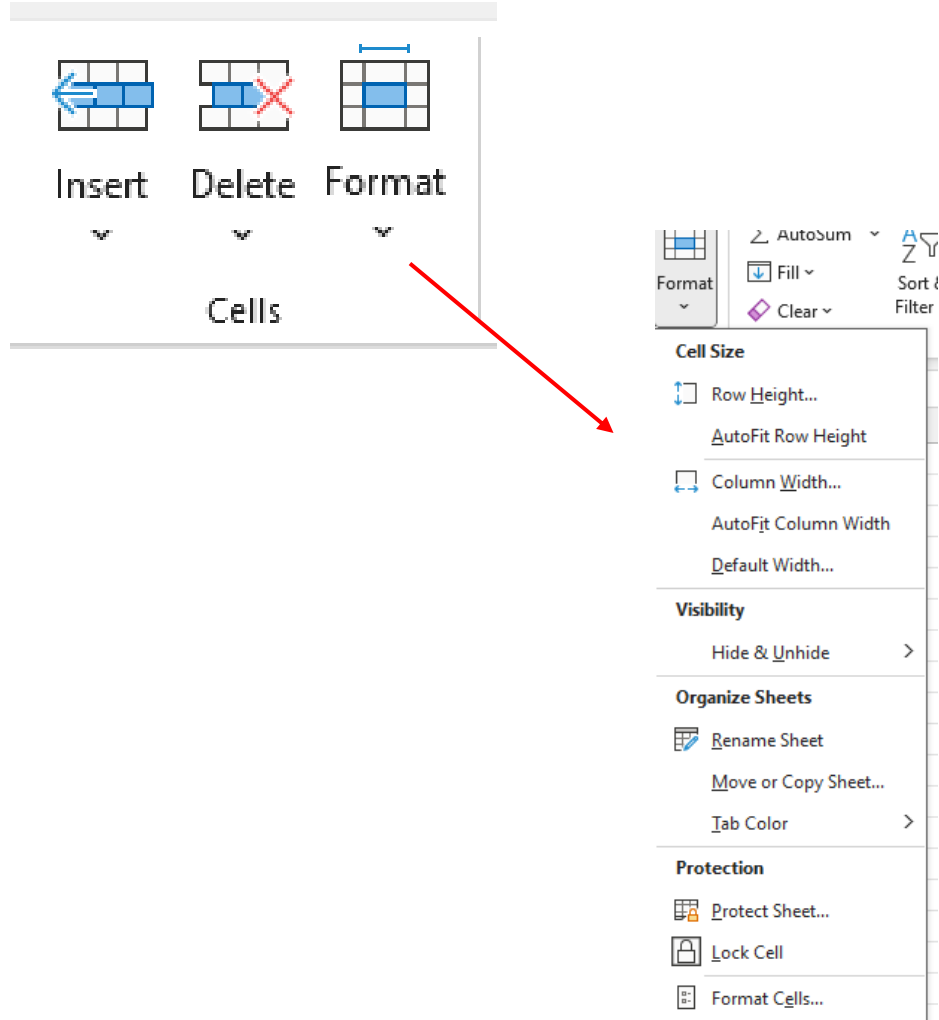
Delete Cells: Removes the selected cells, shifting remaining cells up or to the left.

Delete Sheet Rows: Removes the selected row(s).

Delete Sheet Columns: Removes the selected column(s).

Delete Sheet: Removes the selected worksheet from the workbook.

cells



Format provides options to change the size, visibility, and protection of cells, rows, columns, and sheets.

Row Height: Sets the height of the selected row(s).

AutoFit Row Height: Adjusts the height of the row(s) to fit the contents.

Column Width: Sets the width of the selected column(s).

AutoFit Column Width: Adjusts the width of the column(s) to fit the contents.

Default Width: Sets the default width for all columns in the worksheet.

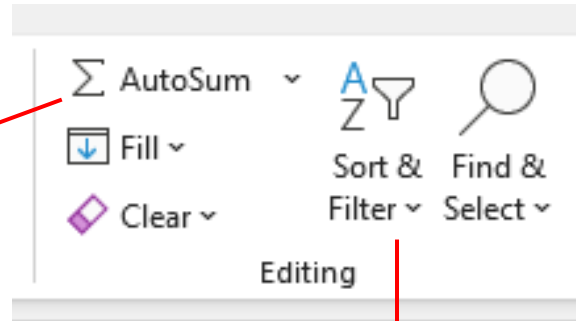
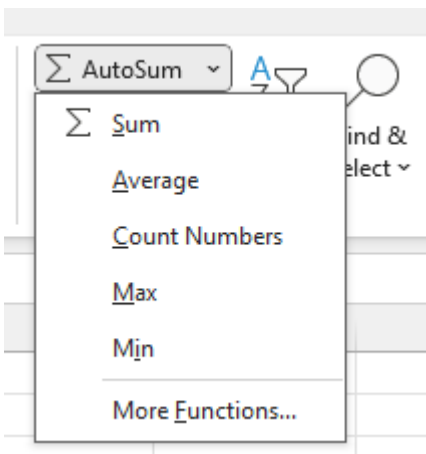
Visibility: Manage the visibility of cells, rows, columns, and sheets.

Organize Sheets: Manage worksheet order and renaming.

Protection: Manage the protection settings of cells and sheets.

Editing

AutoSum quickly adds up a column or row of numbers formula.
Click the drop-down arrow next to the AutoSum button to access other functions like Average, Count Numbers, Max, and Min.

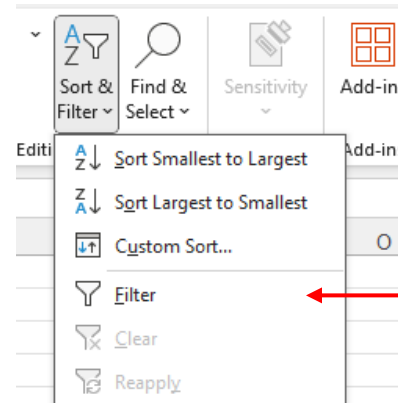


Sort & Filter organizes data and extracts specific information based on criteria.

Sort A to Z: Ascending order.

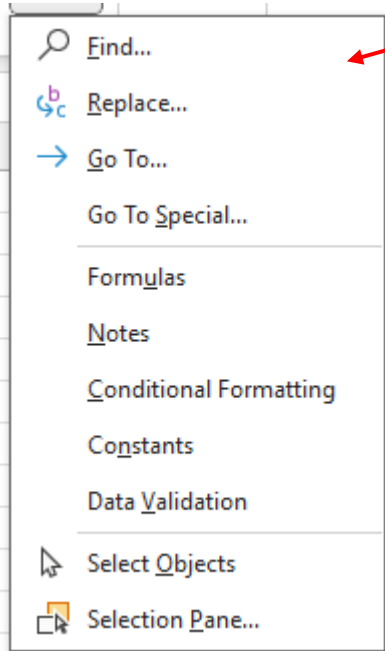
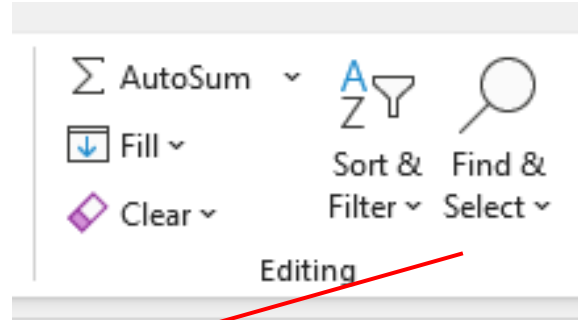
Sort Z to A: Descending order.

Custom Sort: Opens the Sort dialog box for advanced sorting options.



Click the filter dropdown arrow in the column header to apply specific filters.

Editing



Find & Select locates specific text, numbers, or formulas in your worksheet and allows for special selections.

•**Finding Data:**

Choose "Find..." to open the Find and Replace dialog box.

Enter the text or number you want to find and click "Find Next."

•**Replacing Data:**

Choose "Replace..." to open the Find and Replace dialog box.

Enter the text to find and the replacement text, then click "Replace" or "Replace All."

Other Find & Select Options:

Go To: Jump to a specific cell or range.

Go To Special: Select cells based on criteria (e.g., blanks, constants, formulas).

Select Objects: Select graphical objects.

Formulas: Select cells with formulas.

Comments: Select cells with comments.

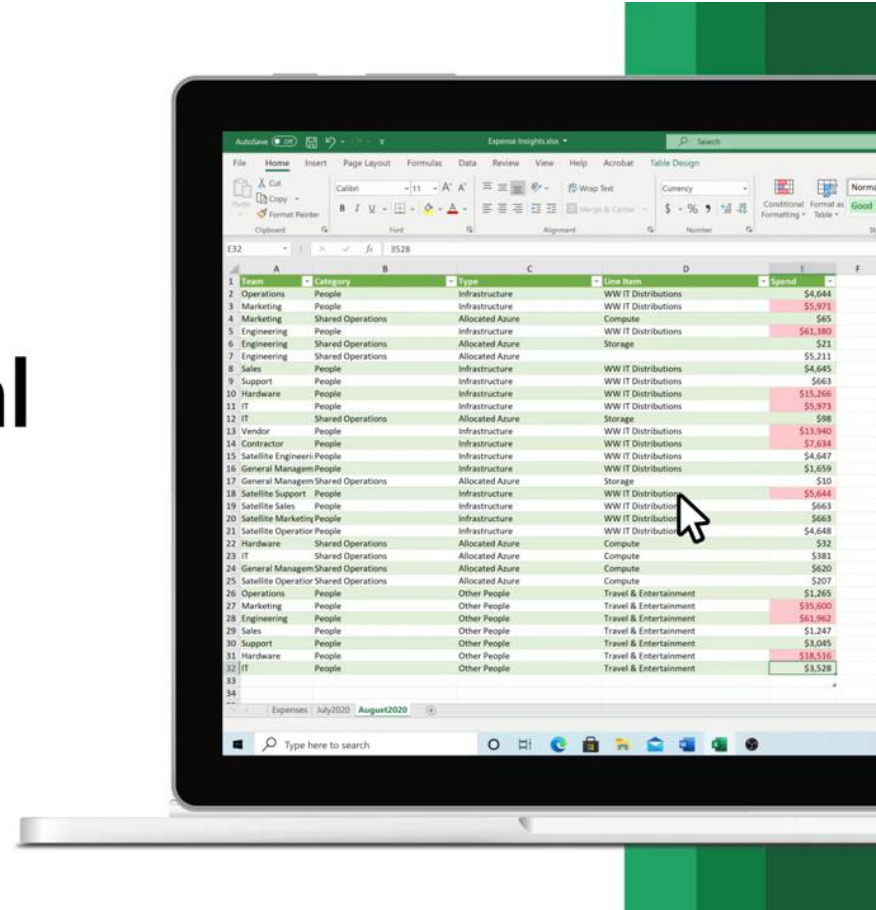
Conditional Formatting: Select cells with conditional formatting.

Data Validation: Select cells with data validation rules.

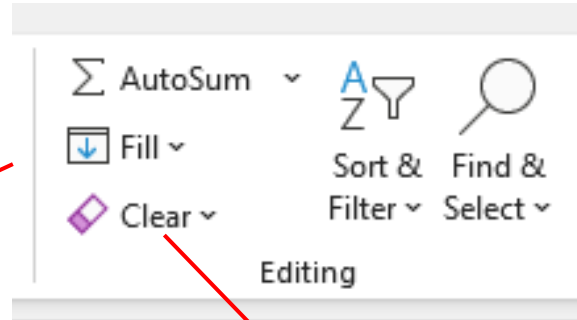
conditional formatting

Microsoft Excel

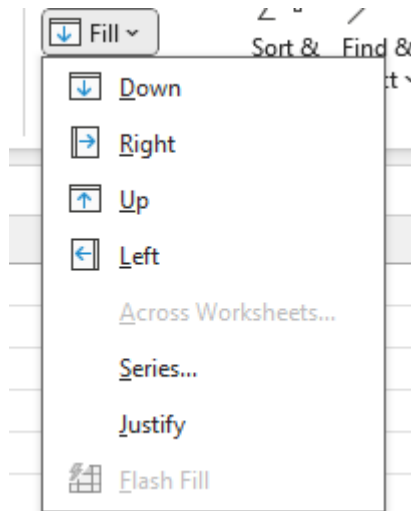
Use conditional formatting



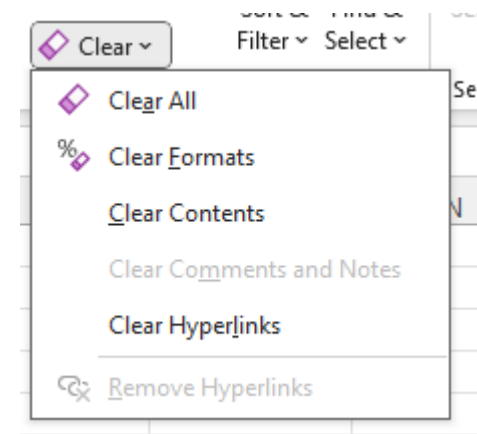
Editing



Fill helps to automatically populate cells with data following a pattern.

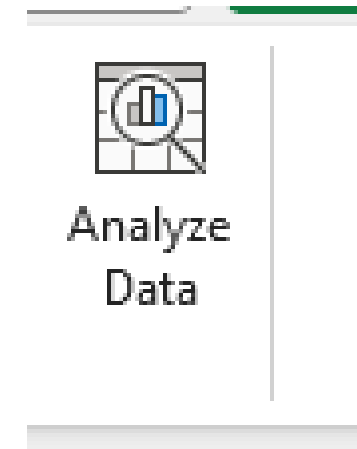
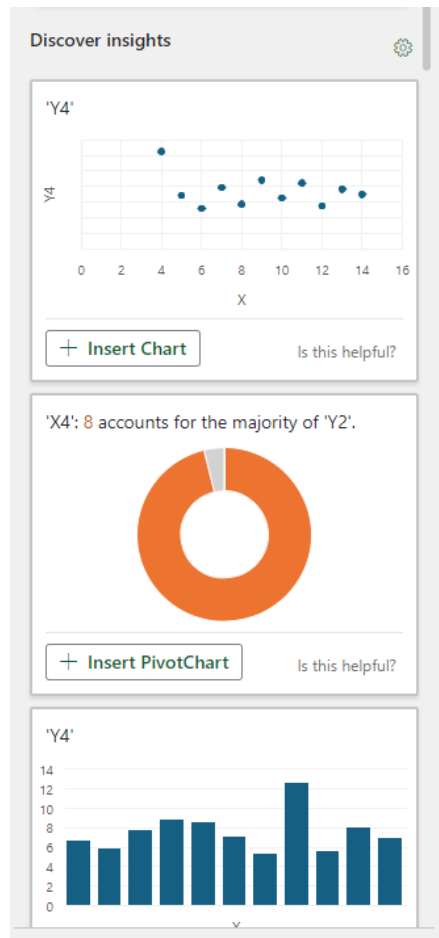


Clear removes content, formats, or comments from selected cells.



Analyze data

help you quickly gain insights from your data, perform complex analyses, and visualize trends. These tools can be found in the **Home**, **Insert**, **Data**, and **Formulas** tabs. Here's a detailed guide to the various Analyze Data tools available in the





We will talk about defining the data types

General : Default format that displays numbers as they are entered, without any specific formatting.

Number: Displays numbers with a specific number of decimal places. You can also include thousand separators.

Currency: Formats numbers as monetary values, including a currency symbol.

Accounting: Like Currency but aligns the currency symbols and decimal points in a column for better readability.

Short Date: Formats dates in a short format (e.g., MM/DD/YYYY).

Long Date: Formats dates in a more detailed format (e.g., Day of Week, Month Day, Year).

Time: Formats numbers as times (e.g., HH ,AM/PM).

Percentage: Converts numbers to a percentage format by multiplying by 100 and adding a percent symbol.

Fraction: Displays numbers as fractions.

Scientific: Formats numbers in scientific notation (e.g., 1.23E+03).

